



How To Build An Intelligent Home Using Raspberry Pi

The proposed Internet of Things (IoT)-based smart system includes such smart devices as smart fridge and smart alarms. Here, the smart fridge will automatically detect daily needs such as eggs and milk stored in it, and if any of it is about to finish, it will automatically send a message to the grocery store. Such smart devices and wireless sensor nodes (WSNs) are connected through a wireless module to a centralised server, Raspberry Pi. The smart fridge uses pressure and temperature sensors for gathering information and sending a message



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The proposed system has three sections: smart fridge, central server and a cellphone. Information is sent and received from the central server through the Internet/cloud. An Android-based app is used for remotely controlling the system. As long as there is Internet coverage, the system can be controlled from anywhere in the world.

Wireless connection reduces difficulty of the setup. The project's block diagram is shown in Fig. 1.

The flowchart in Fig. 2 shows the process of building the system, which includes various sensors, transceivers, microcontroller (MCU) and Raspberry Pi module to control the system.

Smart fridge

The smart fridge detects the actual quantity and availability of grocery items, such as eggs and milk, in the fridge and shows it on the LCD display. For detection, it uses a load cell, which is shown in Fig. 3. When quantity of eggs or milk goes below a particular level, it shows so on the LCD.

When auto mode is on, order is placed automatically through an SMS to the grocery store. When auto mode is off, the user can place the order manually by pushing the buttons provided on the fridge, or using the app. The system then sends the signal to the central server to send the SMS to the grocery shop. Manual switches are shown in Fig. 4.

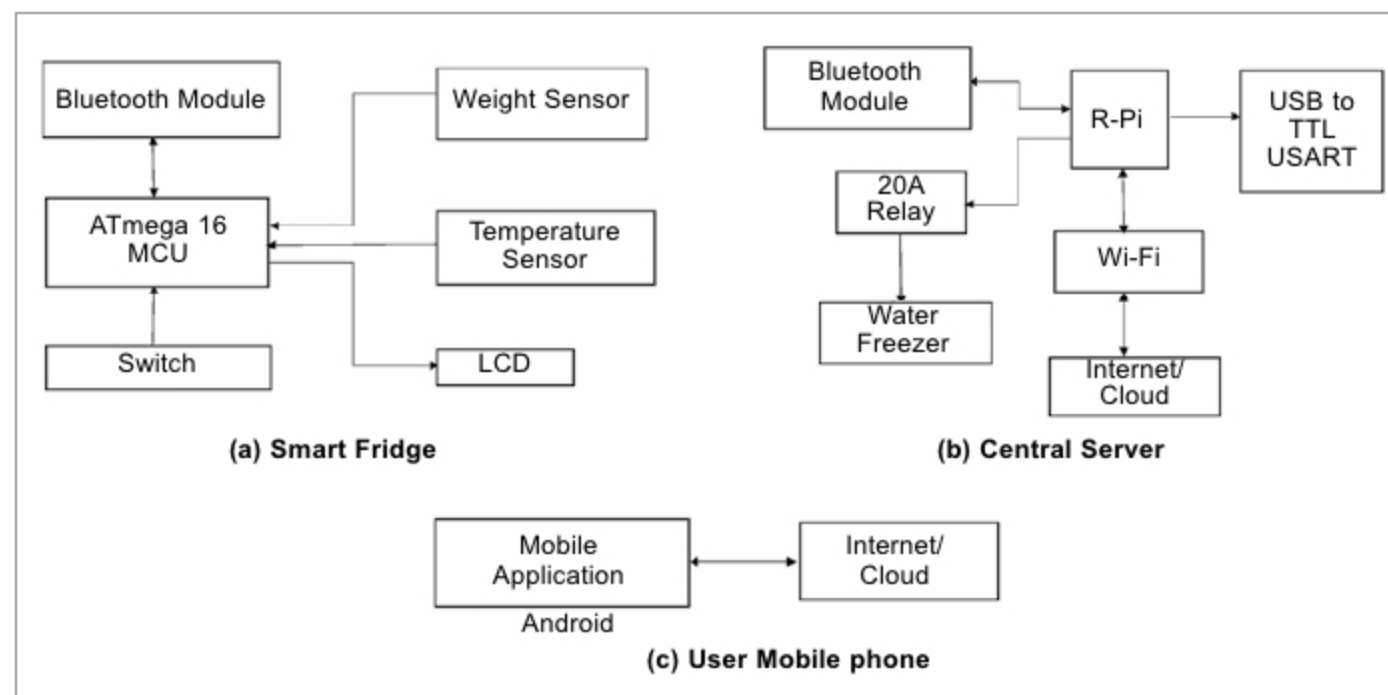


Fig. 1: The intelligent home project